A Project Report on

EatWise:
AI-Based Food Expiry, Usage & Nutrition Optimizer

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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(2025- 2026)

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CERTIFICATE

This is to certify that the project entitled "EatWise: AI- Based Food Expiry, Usage & Nutrition Optimizer " is a Bonafide work of **Aditi Gujar (23106099), Saniya Dhopavkar (23106021), Vaishnavi Bagmar (23106007), Karina Jain (23106097)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Abstract

Food waste and unhealthy eating habits are major challenges in modern society, primarily caused by inefficient food management and lack of awareness regarding nutritional intake. The proposed system, EatWise, is an AI-based smart food management application designed to address these issues by integrating food inventory tracking, expiry monitoring, and personalized dietary recommendations into a single platform. The system enables users to input food items through barcode scanning or manual entry and stores relevant data such as expiry dates, nutritional values, and allergen information in a centralized database.

Using machine learning techniques, the system analyzes food data to predict expiry urgency and prioritize items that should be consumed first, thereby reducing wastage. Additionally, EatWise provides personalized food recommendations based on user-specific goals such as weight loss, weight gain, or maintenance. The system also generates timely alerts for expiring food items and suggests their optimal utilization.

By combining artificial intelligence with everyday food management practices, EatWise promotes efficient resource utilization and supports sustainable consumption. The system aligns with the principles of responsible consumption and production by minimizing food waste while improving dietary awareness. Overall, EatWise serves as a practical and user-friendly solution for enhancing both sustainability and health in daily life.

This project supports Sustainable Development Goal 12 by promoting efficient use of food resources and reducing waste. By tracking inventory and monitoring expiry dates, the system encourages responsible consumption and sustainable resource management aligned with the goals of the United Nations.

Keywords- Artificial Intelligence, Food Waste Reduction, Expiry Prediction, Nutrition Analysis, Smart Food Management, Machine Learning, Sustainable Consumption.

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Chapter 1

Introduction

1.1 Background

Food is one of the most essential resources for human survival, yet a significant portion of it is wasted every day due to inefficient management and lack of awareness. Food waste has become a critical global issue impacting economic resources, environmental sustainability, and food security. A large amount of food produced for consumption is lost at the consumer level due to improper storage, lack of tracking of expiry dates, and inefficient consumption habits. This results in unnecessary wastage of resources such as water, energy, and labor, and contributes to environmental problems including increased carbon emissions and landfill accumulation.

In addition to food wastage, individuals often face challenges in maintaining a balanced and nutritious diet. This is primarily due to a lack of awareness about nutritional values and absence of proper guidance in selecting food based on personal health goals. As lifestyles become more fast-paced, people tend to rely on unstructured eating habits, which further affects their health and well-being.

With advancements in technology, particularly in Artificial Intelligence (AI) and Machine Learning (ML), there is an opportunity to develop intelligent systems that can assist users in managing food more effectively. AI-based systems can analyse data, identify patterns, and provide predictive insights, making them suitable for applications in food tracking and nutrition management. These systems can help in monitoring food expiry, analysing nutritional content, and suggesting optimal consumption strategies.

The proposed system, EatWise, is developed to address these challenges by providing a smart food management solution that integrates food tracking, expiry monitoring, and personalized dietary recommendations into a single platform.

1.2 Motivation

The primary motivation behind the development of EatWise arises from the increasing concern over food wastage and unhealthy eating habits. In many households, food is often discarded due to lack of awareness about its expiry or improper planning of consumption. This not only results in financial loss but also contributes to environmental degradation.

Existing applications available in the market focus on specific functionalities such as calorie tracking or grocery management. However, they fail to provide a comprehensive solution that integrates food inventory management with health-based recommendations. Moreover, these systems do not utilize advanced AI techniques to predict food usage or provide intelligent suggestions.

Another motivating factor is the lack of personalized guidance for users in managing their diet. Individuals have different health goals and dietary requirements, but most applications provide generalized recommendations that may not be suitable for everyone.

EatWise is motivated by the need to create a system that combines multiple functionalities into a single intelligent platform. By integrating AI-based prediction, expiry tracking, and personalized recommendations, the system aims to provide a smarter and more efficient approach to food management while promoting healthier lifestyles.

1.3 Aim and Scope

Aim:

The main aim of this project is to design and develop an AI-based smart food management system that reduces food waste and promotes healthy eating habits through intelligent tracking, analysis, and recommendations.

Scope:

The scope of the EatWise system includes the development of a user-friendly application that allows users to manage their food inventory efficiently. The system enables users to input food items through barcode scanning or manual entry and stores essential details such as expiry dates, nutritional values, and allergen information.

The system also focuses on monitoring food expiry and generating alerts to ensure timely consumption, thereby minimizing wastage. Additionally, it analyses the nutritional content of food items and provides insights to help users make informed dietary choices.

Another important aspect of the system is the implementation of AI-based techniques to predict food usage priority and provide personalized recommendations based on user preferences and health goals. The scope of the project is primarily limited to household-level food management but can be extended to commercial applications in the future.

1.4 Relevance to Sustainable Development Goals

The EatWise system is closely aligned with **United Nations Sustainable Development Goal 12: Responsible Consumption and Production**, which focuses on reducing waste generation and promoting sustainable use of resources.

Food waste is a major contributor to environmental degradation and inefficient resource utilization. By enabling users to track food items, monitor expiry dates, and prioritize consumption, EatWise helps reduce food wastage at the consumer level. This directly supports the objective of minimizing waste and ensuring efficient utilization of available resources. In addition, the system promotes awareness about responsible consumption by encouraging users to make better food choices and utilize food effectively. It also contributes to environmental sustainability by reducing the carbon footprint associated with food production, transportation, and disposal.

Furthermore, EatWise indirectly supports other sustainable development goals such as good health and well-being by promoting healthier eating habits through nutritional analysis and personalized recommendations.

Thus, the system not only addresses individual food management challenges but also contributes to broader global sustainability objectives.

Chapter 2

Literature Survey

2.1 Literature Survey

In recent years, rapid advancements in mobile technology and Artificial Intelligence (AI) have significantly influenced the development of applications aimed at improving food management, nutrition tracking, and dietary planning. Various systems have been introduced to assist users in monitoring their food intake, maintaining a balanced diet, and making healthier lifestyle choices. Nutrition tracking applications such as MyFitnessPal and similar platforms allow users to log daily food intake, monitor calorie consumption, and track macronutrient distribution, thereby supporting fitness and health goals [1]. These systems have gained widespread adoption due to their ease of use and accessibility. However, they rely heavily on manual data entry, which can be time-consuming and prone to human error. Furthermore, research has shown that nutritional databases used in such applications may contain inconsistencies and inaccuracies, which can affect the reliability of dietary analysis [2]. Additionally, these systems focus primarily on calorie tracking and do not provide features for food inventory management or expiry tracking, which are essential for reducing food waste.

Food scanning and rating applications such as Yuka utilize barcode scanning technology to provide users with detailed information about food products, including ingredient composition, nutritional values, and health scores [3]. These applications aim to increase awareness and enable users to make informed purchasing decisions. Studies indicate that such systems can positively influence consumer behavior by encouraging healthier food choices and improving nutritional awareness [4]. Despite these advantages, food scanning applications have certain limitations. They do not consider individual user preferences, dietary restrictions, or health goals, and their recommendations are often generalized. Moreover, these systems lack the ability to track food inventory or monitor expiry dates, limiting their effectiveness in addressing food wastage issues.

Inventory management applications such as Out of Milk are designed to help users organize grocery lists and manage pantry items efficiently [5]. These systems provide features for tracking purchased items and planning shopping activities, thereby improving organization and reducing redundancy in purchases. However, they do not include advanced functionalities such as nutritional analysis, expiry tracking, or intelligent recommendations. As a result, they are not effective in promoting healthy eating habits or reducing food waste. The absence of AI-based decision-making further limits their ability to provide meaningful insights to users.

With the advancement of AI and computer vision technologies, several research efforts have focused on developing food recognition systems. Systems such as NutrifyAI use deep learning algorithms to identify food items from images and provide real-time nutritional analysis [6]. These systems enhance user convenience by reducing the need for manual input and enabling automatic data collection. However, they are primarily focused on food identification and do not integrate features such as food inventory tracking or expiry monitoring. Their performance may also be affected by factors such as image quality and dataset limitations, which can impact accuracy.

AI-based recommendation systems have also gained attention in the field of food and nutrition. Systems such as Yum-me provide personalized meal recommendations based on user preferences, dietary requirements, and health goals [7]. These systems enhance user experience by offering tailored suggestions that align with individual needs. However, they do not consider real-time food availability or inventory data when generating recommendations. As a result, suggested meals may not always be practical or feasible, particularly when required ingredients are unavailable or nearing expiry. This limits their effectiveness in reducing food waste and optimizing resource utilization.

In addition to individual applications, research has been conducted on smart food waste management systems that utilize data analytics and AI techniques to predict and reduce food wastage [8]. These systems analyze consumption patterns and provide insights for better resource management, contributing to sustainability and environmental conservation. However, many of these solutions are designed for large-scale environments such as restaurants, supply chains, or commercial establishments and are not suitable for individual household use. They often lack user-friendly interfaces and do not provide personalized recommendations, which restricts their adoption among general users.

Overall, the literature survey reveals that existing systems address specific aspects of food management, such as nutrition tracking, food scanning, inventory management, or recommendation systems, but fail to provide a comprehensive and integrated solution. Most systems operate independently and lack coordination between different functionalities. There is a clear absence of systems that combine food inventory tracking, expiry monitoring, nutritional analysis, and AI-based recommendation into a single platform. Furthermore, many applications depend on manual input, lack personalization, and do not focus on reducing food waste at the consumer level. These limitations highlight a significant research gap and emphasize the need for an intelligent and integrated solution like EatWise. The proposed system aims to address these challenges by combining multiple functionalities into a unified platform, enabling efficient food management, reducing waste, and promoting healthier and more sustainable consumption practices.

Table 1: Literature Survey

Paper	Key Features	Advantages	Limitations
MyFitnessPal	Calorie tracking, diet monitoring	Helps maintain fitness goals	Manual input, no expiry tracking
Nutrition Data Studies	Nutrition database analysis	Improves data awareness	Inconsistent data
Yuka	Barcode scanning, food rating	Encourages healthy choices	No personalization, no tracking
Consumer Behavior Studies	Food awareness impact	Improves decision-making	No practical implementation
Out of Milk	Grocery and pantry management	Organizes food items	No nutrition or AI
NutrifyAI	AI food recognition	Real-time analysis	No expiry tracking
Yum- me	Food recommendation system	Personalized suggestions	No inventory tracking
Smart Food Waste Systems	Waste reduction models	Focus on sustainability	Limited user interaction
Onyeaka et al.	AI for food waste reduction and supply chain optimization	Improves efficiency, supports sustainability	No user-level system, mostly theoretical
Onyeaka et al. (2025)	AI-based food spoilage prediction and monitoring	Better prediction, early detection	Lacks personalization, not for households

Chapter 3

Limitation of Existing system

3.1 Limitations of Existing system

Existing food management and nutrition-related systems have improved awareness regarding dietary habits and food consumption; however, they still face several limitations that reduce their effectiveness in solving real-world problems such as food wastage and efficient food utilization. Most existing systems are designed to perform specific functions such as calorie tracking, food scanning, or grocery management, but they lack integration of these features into a single platform. This fragmented approach forces users to rely on multiple applications, making the process inefficient and less user-friendly.

One of the major limitations of current systems is the absence of proper food expiry tracking. Many applications do not provide mechanisms to monitor the shelf life of food items or generate timely alerts before they expire. As a result, users often forget about stored food and fail to consume it within the required time, leading to unnecessary wastage. In addition, these systems lack real-time notifications and intelligent reminders, which further reduces their ability to minimize food waste. Another important limitation is the lack of Artificial Intelligence-based prediction and analysis. Most systems operate on static data and do not analyze user consumption patterns. Without predictive capabilities, these systems cannot prioritize food items based on urgency or suggest which items should be consumed first. This reduces their effectiveness in guiding users toward efficient food utilization.

Furthermore, many applications provide generalized recommendations and lack personalization. Users have different dietary needs, preferences, and health goals, but existing systems fail to provide customized suggestions. This limits their usefulness in helping users maintain a healthy and balanced diet. Additionally, most applications depend heavily on manual data entry, which is time-consuming and prone to errors. This can lead to inaccurate information and reduced reliability of the system.

Another challenge is the inconsistency of nutritional data across different platforms. Many applications rely on large databases that may contain outdated or incorrect values, leading to inaccurate analysis and misleading results. Moreover, existing systems are not specifically designed to reduce food waste, as they do not provide actionable insights such as prioritizing near-expiry items or suggesting ways to utilize them efficiently.

In addition, several advanced food waste management systems are designed for large-scale applications such as restaurants and supply chains, making them complex and unsuitable for individual users. These systems lack user-friendly interfaces and are not easily accessible for household use.

Research Gap:

Lack of an integrated system that combines food tracking, expiry monitoring, nutrition analysis, and recommendations in a single platform

• Absence of intelligent systems that utilize Artificial Intelligence for predictive analysis and decision-making

• Limited availability of personalized dietary recommendations based on user preferences and health conditions

• No effective mechanisms in existing systems to reduce food waste at the consumer or household level

• Lack of real-time alerts and prioritization of food items based on expiry dates

• Existing applications focus on isolated functionalities rather than a comprehensive solution

• Minimal support for sustainable consumption practices and awareness

• Need for a smart, efficient, and user-friendly system like EatWise to address these gaps.

The proposed improvement introduces an AI-based system with expiry tracking to improve food management and reduce waste. It provides a reliable and integrated solution that combines inventory management with artificial intelligence. The system also offers personalized recommendations based on user preferences and available inventory. By integrating tracking and prediction features, it helps monitor food freshness and forecast potential waste. Additionally, the system is designed to be user-friendly and can be integrated into real-world environments, making food management more efficient and practical.

Table 2: Limitations of existing systems

Aspect	Existing System features	Limitations	Improvement in proposed system
Integration	Separate apps for tracking, scanning, inventory	No unified system	All features integrated in one platform
Expiry Tracking	Mostly absent	No alerts for expiring food	Real-time expiry alerts
AI Prediction	Minimal or no AI usage	No food prioritization	AI-based prediction and prioritization
Personalization	General recommendations	No user-specific suggestions	Personalized diet recommendations
Data Entry	Manual input required	Time-consuming, error-prone	Barcode scanning + automation
Nutritional Data	Large databases	Inconsistent/inaccurate data	Reliable and structured data
Food waste Reduction	Not a focus	No actionable insights	Waste reduction through smart alerts
Usability	Complex or fragmented systems	Not user-friendly	Simple, integrated interface

Chapter 4
Problem Statement and Objective

4.1 Problem Statement

Food wastage has become a significant issue due to inefficient food management practices, lack of awareness about expiry dates, and improper consumption habits. In many households, food items are often forgotten or not consumed within their shelf life, leading to unnecessary waste of resources. At the same time, individuals face difficulty in maintaining a balanced and healthy diet due to limited knowledge of nutritional values and lack of proper guidance.

Existing systems and applications focus on isolated functionalities such as calorie tracking, food scanning, or grocery management, but they fail to provide an integrated solution that combines food inventory tracking, expiry monitoring, and personalized dietary recommendations. Additionally, these systems lack advanced Artificial Intelligence capabilities to predict food usage or prioritize items based on urgency, which reduces their effectiveness in real-world scenarios.

Moreover, the absence of real-time notifications and intelligent alert mechanisms further contributes to inefficient food utilization. Users often struggle to keep track of multiple food items, especially in busy and fast-paced lifestyles, which increases the chances of food spoilage and wastage. In many cases, users are unaware of which items need immediate consumption, resulting in poor prioritization and resource mismanagement.

Another critical issue is the heavy reliance on manual tracking methods, such as remembering expiry dates or maintaining physical notes, which are unreliable and prone to human error. This lack of automation makes it difficult for users to consistently manage food inventory. Furthermore, existing systems do not provide meaningful insights or actionable suggestions that can guide users toward better food consumption practices.

In addition, there is limited awareness among users regarding sustainable consumption and the environmental impact of food wastage. Food waste not only leads to economic loss but also contributes to environmental problems such as increased carbon footprint and resource depletion. Despite the growing importance of sustainability, most available applications do not actively promote or support eco-friendly practices.

The lack of personalization is another major challenge, as users have diverse dietary preferences, health conditions, and lifestyle requirements. Current systems fail to adapt to individual needs, resulting in generic recommendations that may not be useful or effective. This reduces user engagement and limits the overall impact of such applications.

Therefore, there is a need for an intelligent, integrated, and user-friendly system that can efficiently manage food resources, provide real-time alerts, offer personalized recommendations, and promote sustainable consumption practices. Such a system should leverage modern technologies to simplify food management, reduce wastage, and assist users in maintaining healthier and more responsible lifestyles.

4.2 Objectives

The primary objective of this project is to develop an intelligent food management system that helps in reducing food waste and promoting healthy eating habits. The system aims to provide an efficient solution for tracking food items and monitoring their expiry dates, ensuring timely consumption and minimizing wastage.

Another objective is to analyze the nutritional content of food items and provide meaningful insights to users, enabling them to make informed dietary decisions. The system also focuses on delivering personalized recommendations based on user preferences and health goals such as weight loss, weight gain, or maintenance.

Chapter 5

Proposed System and Methodology

5.1 System Overview

The EatWise system is an intelligent food management application designed to help users efficiently manage food inventory, reduce wastage, and maintain healthy eating habits. The system provides a unified platform that integrates multiple functionalities such as food tracking, expiry monitoring, nutritional analysis, and basic recommendation mechanisms. It is developed using a cross-platform frontend and a cloud-based backend to ensure accessibility, scalability, and real-time data synchronization.

The system allows users to add food items either manually or through barcode scanning, making data entry convenient and efficient. Each food item is stored with details such as name, quantity, and expiry date. The system continuously monitors this data and generates alerts for items that are nearing expiration. This helps users prioritize consumption and avoid unnecessary wastage.

Additionally, the system provides basic nutritional insights and suggests food items that should be consumed first based on their expiry dates. The integration of frontend and backend components ensures seamless communication and efficient data processing. Overall, the EatWise system offers a simple, user-friendly, and effective solution for smart food management.

5.2 System Architecture

The architecture of the EatWise system follows a client-server model, consisting of three main components: the frontend (client side), the backend (server side), and the database. This layered architecture ensures efficient data handling, scalability, and smooth interaction between system components.

The frontend is developed using Flutter and serves as the user interface of the application. It is responsible for capturing user inputs, displaying food inventory data, and presenting alerts and recommendations. The frontend communicates with the backend through REST APIs to send and receive data in real time.

The backend is implemented using Supabase, which provides authentication, database management, and API services. It handles core functionalities such as data processing, food item storage, expiry monitoring, and user management. The backend ensures secure data handling and efficient execution of system operations.

The database stores all relevant information, including user details, food inventory, expiry dates, and nutritional data. It enables quick retrieval and updating of data, ensuring that users always have access to the latest information.

The overall workflow begins with user input from the frontend, which is sent to the backend for processing and storage in the database. The backend then analyzes the data and sends appropriate responses, such as alerts or recommendations, back to the frontend. This continuous interaction ensures that the system operates efficiently and provides real-time assistance to users.

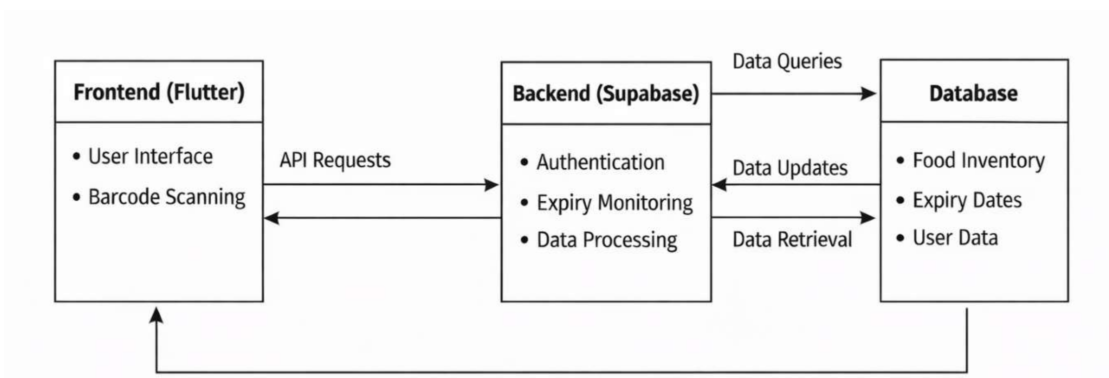

Figure 1: System Architecture

5.3 Methodology

The methodology of the EatWise system follows a structured approach to efficiently manage food inventory, reduce food wastage, and provide useful recommendations to users. The system operates through a sequence of well-defined steps, ensuring smooth data flow and accurate processing of information.

The process begins with data input, where users add food items to the system either manually or by using barcode scanning. The input includes essential details such as food name, quantity, and expiry date. This step ensures that all relevant information is captured for further processing.

Once the data is entered, it is securely stored in the database using the backend system. The database maintains all user-related information and food item details, allowing easy access and updates whenever required. This ensures data consistency and reliability throughout the system.

The next step involves data processing and monitoring, where the system continuously checks stored food items and tracks their expiry dates. The system applies predefined logic to identify items that are close to expiration. This real-time monitoring helps in maintaining an updated status of all food items.

Following this, the system performs analysis and prioritization of food items. Based on expiry dates, the system determines which items need to be consumed first. Basic intelligent logic is applied to prioritize food items and assist users in making better decisions regarding food consumption.

After analysis, the system generates alerts and recommendations. Notifications are provided to users for food items nearing expiry, and suggestions are given to consume specific items first. This step helps in reducing food wastage and improving efficiency in food utilization.

Finally, the system provides the output in the form of a user-friendly interface, where users can view their food inventory, receive alerts, and access recommendations. The continuous interaction between frontend and backend ensures real-time updates and smooth system performance.

Overall, the methodology of EatWise follows a logical flow of Input → Storage → Processing → Analysis → Output, ensuring an effective and intelligent approach to food management.

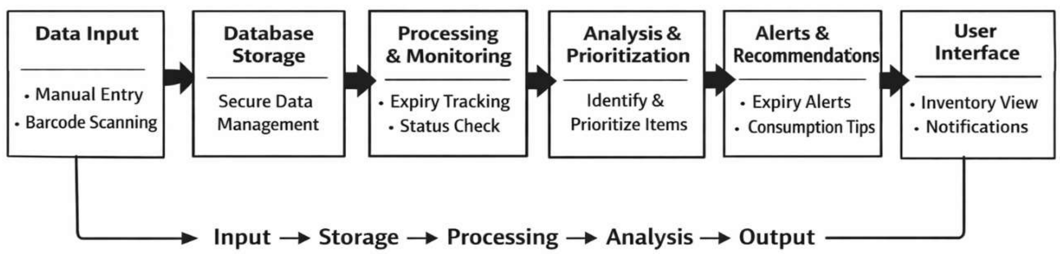

Figure 2: Methodology

Chapter 6

Experimental Setup

6.1 Frontend Setup

The frontend of the proposed EatWise system is developed using modern web technologies to provide an interactive and user-friendly interface. It is designed to ensure smooth navigation and efficient interaction between the user and the system. The frontend is responsible for capturing user input, displaying food inventory details, showing expiry alerts, and presenting nutritional insights in a clear and organized manner.

The system implements a responsive design approach, ensuring compatibility across different devices such as desktops and mobile phones. Users can add food items through manual entry or barcode scanning, and the interface dynamically updates based on backend responses. The frontend also manages API communication with the backend to fetch and display real-time data. Proper validation techniques are applied to ensure accurate data entry and improve overall usability.

6.1.1 Frontend Technologies Used

Flutter – Cross-platform UI framework for building mobile applications
Visual Studio Code (VS Code) – Code editor for application development
Barcode Scanner (Flutter Plugin) – For scanning food product barcodes
Android Emulator / Physical Device – Testing the mobile application
Web Browser – Used for testing and demo purposes

6.2 Backend Setup

The backend of the proposed EatWise system is developed using Node.js with the Express.js framework, which provides a scalable and efficient environment for building RESTful APIs. The backend is responsible for handling data processing, managing food inventory, monitoring expiry dates, and ensuring communication with the frontend. The system follows a structured workflow where user inputs are processed, stored, and analyzed to generate meaningful outputs.

The backend manages user authentication, stores food item details such as name, quantity, and expiry date, and continuously monitors stored data to identify items nearing expiration. It also handles core functionalities such as generating alerts, analyzing nutritional information, and providing personalized recommendations. Additionally, basic AI-based logic is implemented to prioritize food items based on urgency and consumption patterns, ensuring efficient food utilization and reducing food waste.

6.2.1 Backend Technologies used

Supabase – Backend-as-a-Service platform
Supabase Auth – User authentication and session management
Supabase Database – Storage of user details, food items, and expiry data
REST APIs – Communication between frontend and backend
Cloud Storage (Supabase) – Secure data storage and management

Chapter 7

Results and Discussion

The EatWise system was successfully implemented and tested to evaluate its performance in managing food inventory, monitoring expiry dates, and providing user-friendly recommendations. The system demonstrated effective functionality in storing food item details, including name, quantity, and expiry date, and retrieving this information in real time. The integration between the frontend and backend ensured smooth data flow and efficient communication, allowing users to interact seamlessly with the application.

One of the key outcomes observed during testing was the system's ability to generate timely alerts for food items nearing their expiry. This feature proved to be highly useful in helping users identify and prioritize items that need immediate consumption, thereby reducing food wastage. The alert mechanism functioned accurately based on the stored expiry data and provided notifications in a clear and understandable manner.

The barcode scanning functionality was also tested and found to be efficient in reducing manual effort required for data entry. Users were able to quickly add food items by scanning product barcodes, which improved the overall usability and convenience of the system. In cases where barcode scanning was not available, manual entry options ensured flexibility and accessibility.

The system also provided basic nutritional insights based on stored data, enabling users to make informed decisions about their diet. Although the current implementation uses simple logic for recommendations, it effectively demonstrated the potential for integrating more

advanced AI-based features in the future. The prioritization of food items based on expiry dates further enhanced decision-making by guiding users on which items to consume first.
From a performance perspective, the application showed stable behavior during testing, with minimal latency in data retrieval and processing. The use of cloud-based backend services ensured secure data storage and efficient handling of user information. The user interface was responsive and easy to navigate, contributing to a positive user experience.
However, certain limitations were observed during testing. The accuracy of recommendations is currently limited due to the use of basic logic instead of advanced machine learning models. Additionally, the system relies on user input for food data, which may lead to inaccuracies if incorrect information is entered. Barcode scanning functionality may also depend on the availability of product data in the database.
Overall, the results indicate that the EatWise system effectively addresses the problem of food management at the household level. It successfully integrates key features such as food tracking, expiry monitoring, and basic recommendations into a single platform.

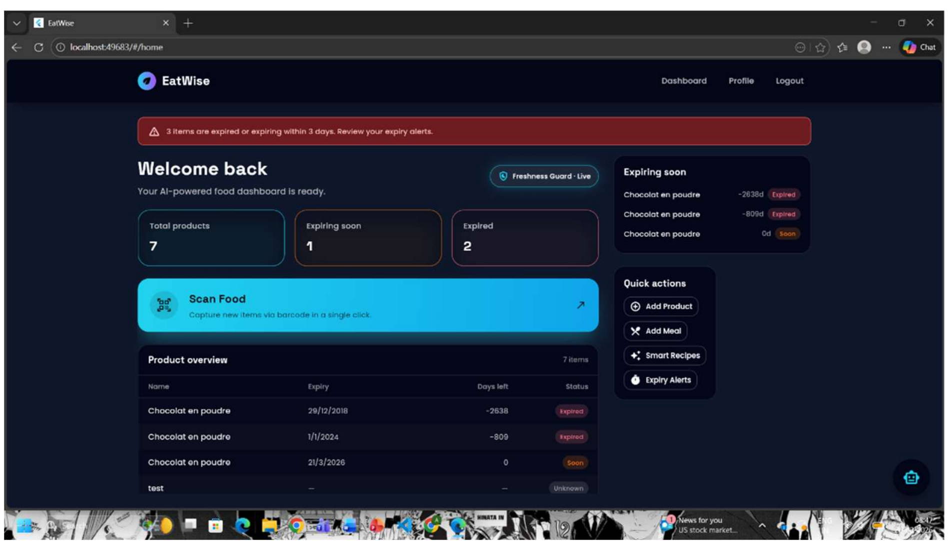

Figure 3: Login Page - Enables users to securely sign in to access their personalized food inventory and features.

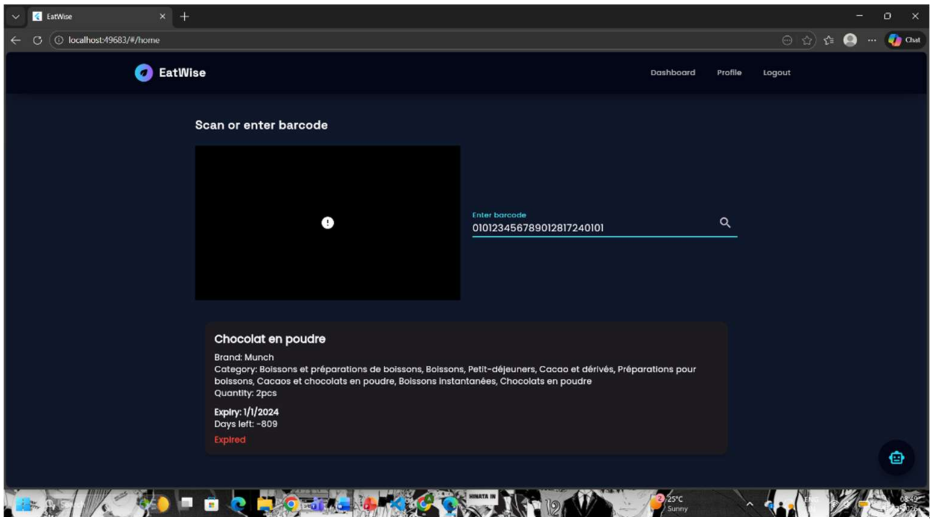

Figure 4: Barcode Entry Page - Allows users to scan or enter product barcodes to automatically add items and their details to the inventory.

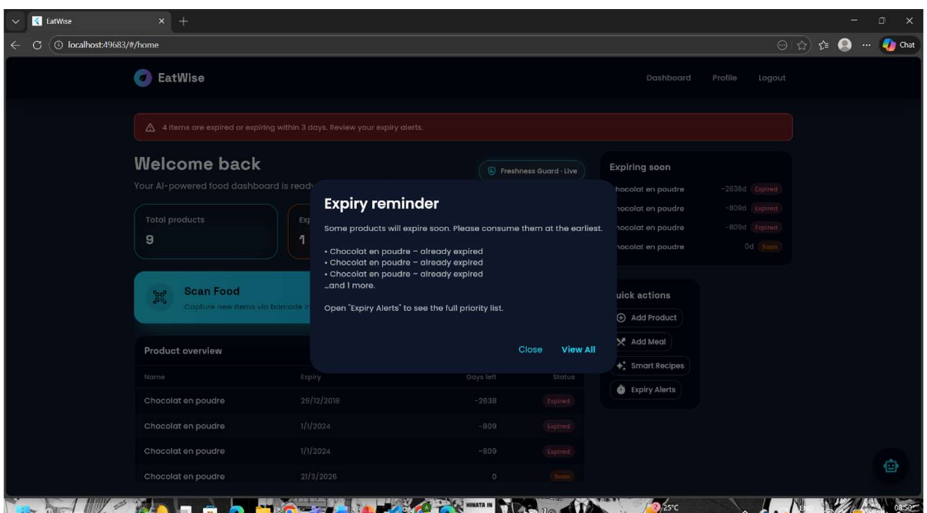

Figure 5: Expiry Reminder - Notifies users about upcoming and overdue product expirations to help prevent food waste and ensure timely usage.

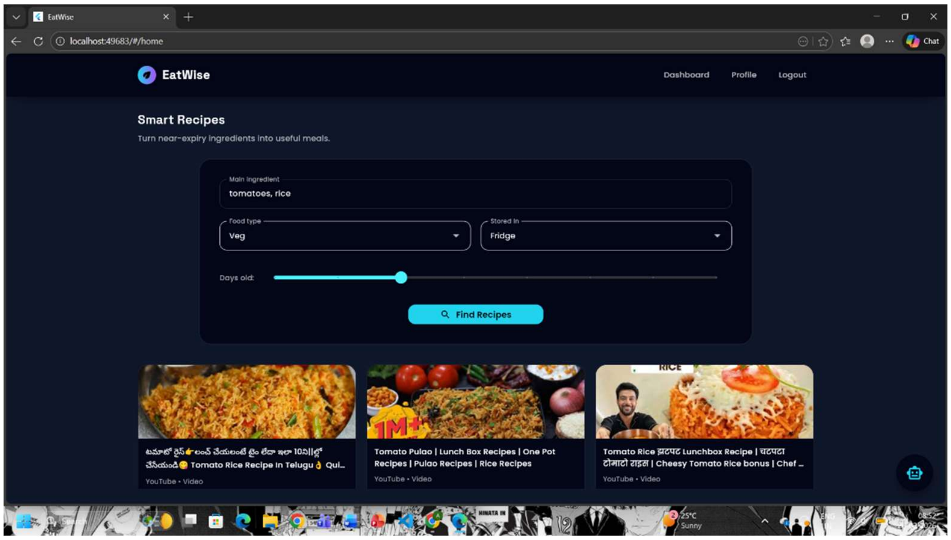

Figure 6: Recipe Recommendation Page - Takes leftover food as input and redirects users to relevant recipe videos for easy preparation.

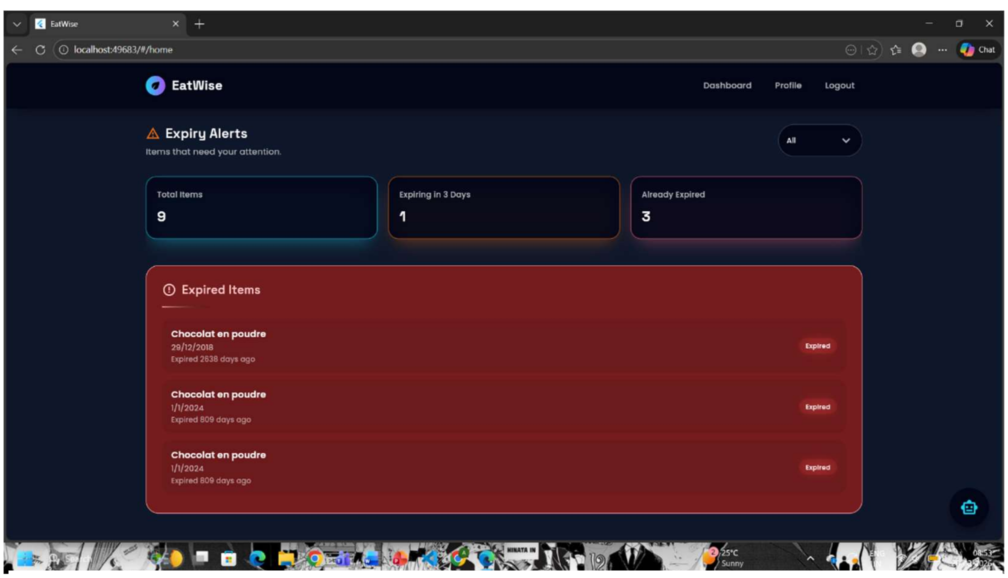

Figure 7: Expiry Alerts Page - Displays expiry alerts with total items, soon-to-expire items, and a list of already expired products.

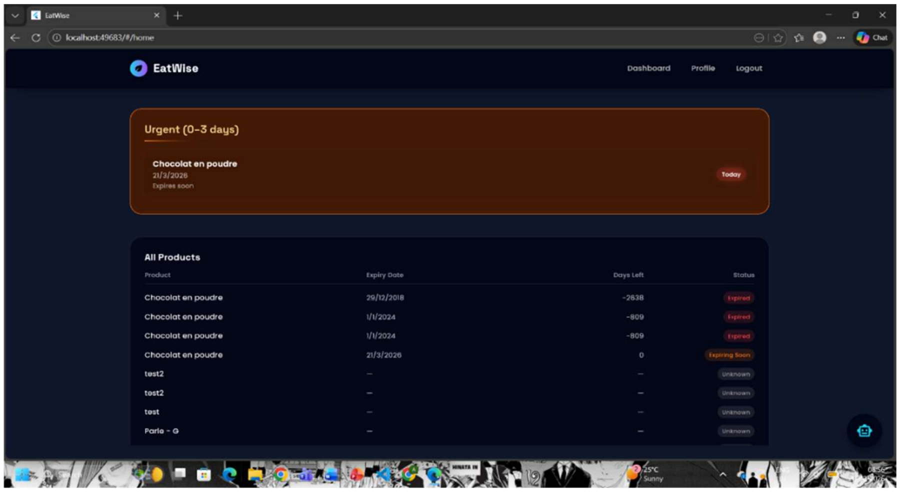

Figure 8: Food Prioritization Page - Shows urgent items and a detailed table of all products with expiry dates, days left, and status labels.

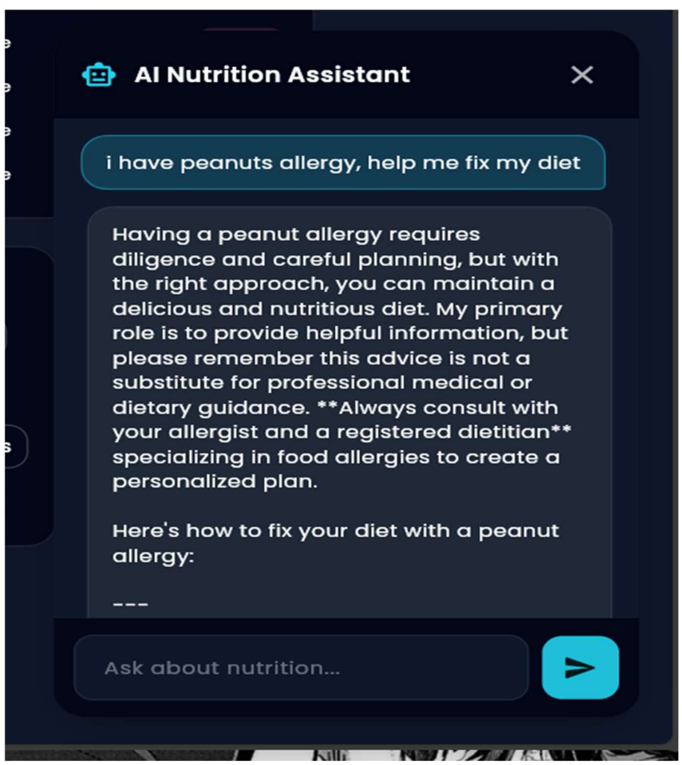

Figure 9: AI Nutritionist Chatbot - hat interface providing dietary advice based on user input about food allergies (e.g., peanut allergy).

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The EatWise system has been successfully designed and developed as an intelligent food management solution that aims to reduce food waste and promote healthy eating habits. The system integrates multiple functionalities such as food inventory tracking, expiry monitoring, and nutritional analysis into a single user-friendly platform. By leveraging modern technologies, the application provides an efficient way for users to manage their food resources and make informed dietary decisions.

One of the key contributions of the system is its ability to track food items and generate timely alerts for products nearing their expiry. This helps users utilize food efficiently and minimizes unnecessary wastage. Additionally, the system supports barcode scanning and manual entry, making it convenient for users to add and manage food items. The integration of backend services ensures secure storage and seamless data management, while the frontend provides an intuitive interface for smooth user interaction.

Furthermore, the system incorporates basic intelligent logic to prioritize food items based on urgency and provide meaningful recommendations. This enhances decision-making and encourages users to adopt sustainable consumption practices. Overall, EatWise addresses the limitations of existing systems by offering a comprehensive and integrated approach to food management. It not only improves efficiency but also contributes to environmental sustainability and better health outcomes.

8.2 Future Scope

- Implementation of image-based food recognition using camera input to identify food items automatically.
- Support for multi-user environments such as families or shared households for collaborative food management.
- Expansion of the system for commercial use in restaurants, grocery stores, and supply chain management.
- Integration with online grocery platforms for automated shopping suggestions based on current inventory.
- Development of a comprehensive smart food ecosystem combining food management, health monitoring, and sustainability practices.

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